Monte Carlo Simulation for Financial Decision-Making

Investment Withdrawal Strategies and Insurance Loss-Ratio Risk

Portfolio Project Report

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Abstract

This report develops and evaluates two spreadsheet-based Monte Carlo simulation models for **financial decisions** under uncertainty. **The first model** studies the longevity of a $1,000,000 **investment fund** split equally between stocks and bonds under two withdrawal policies: **a parallel policy** that draws $60,000 from each asset class annually and **a bond-first policy** that draws from bonds before stocks. **The second model** evaluates **annual insurance loss-ratio risk** when policy volume, liability coverage, claim occurrence, and claim severity are stochastic.

Rather than relying on single deterministic forecasts, both models generate empirical outcome distributions and evaluate means, dispersion, confidence intervals, tail probabilities, and convergence. Across 2,000 investment paths, the parallel and bond-first policies produce mean fund lives of 11.9445 and 12.5695 years, respectively, a difference of 0.6250 years. In the insurance simulation, 500 simulated years produce a mean loss ratio of 0.8367, a standard deviation of 0.8905, and an estimated probability of 29.4% that the annual loss ratio exceeds one.

The results illustrate two complementary uses of simulation: comparing dynamic policies whose consequences accumulate over time, and characterizing aggregate risk generated by multiple interacting random variables.

Keywords: Monte Carlo simulation; geometric Brownian motion; stopping time; investment strategy; insurance risk; loss ratio; convergence; spreadsheet modeling

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1. Introduction

Many **financial decisions** are difficult not because their accounting identities are complex, but because outcomes depend on sequences of uncertain events. A withdrawal rule that appears equivalent in a one-period comparison can alter the amount of capital left exposed to future returns. Likewise, two insurance environments can have similar expected losses while generating very different annual volatility and tail exposure. Monte Carlo simulation provides a natural framework for such problems because it represents uncertainty explicitly, propagates it through the model, and produces an empirical distribution of decision-relevant outcomes rather than a single point forecast.

The project contains two related simulation studies. The **investment model** is a dynamic stochastic system: annual stock returns evolve randomly, withdrawals change the state of the portfolio, and the principal output is the stopping time at which the fund is depleted. The **insurance model** is an aggregate-risk system: the number of policies, coverage amount, excess-coverage purchase, claim occurrence, and claim severity jointly determine annual total premiums, total losses, and the loss ratio. Together, the studies demonstrate how spreadsheet simulation can support policy comparison, risk measurement, and managerial interpretation.

The report is organized as a compact empirical paper. Section 2 describes the simulation framework and output-analysis principles. Sections 3 and 4 present the investment and insurance models, respectively, including assumptions, implementation, results, convergence, and implications. Section 5 synthesizes the analytical lessons and limitations, and Section 6 concludes.

2. Simulation Framework and Analytical Method

2.1 Monte Carlo simulation and empirical output distributions

Monte Carlo simulation treats uncertain model inputs as random variables and repeatedly samples realizations from their assumed distributions. Each replication produces an outcome from the modeled system; repeated replications form an empirical approximation to the output distribution. In spreadsheet implementations, uniform pseudo-random draws generated by RAND() can be transformed into Normal, Bernoulli, Uniform, or Exponential random variables through inverse-distribution or indicator formulations. This approach keeps the stochastic logic transparent and auditable within the workbook.

For an output variable X observed across N independent simulation replications, the project summarizes the simulated distribution using the sample mean, sample standard deviation, confidence intervals for the mean, observed extrema, and event probabilities. These statistics serve different purposes: the mean represents central tendency, the standard deviation represents dispersion, a confidence interval quantifies Monte Carlo estimation precision, and an event probability directly measures the frequency of a decision-relevant threshold event.

$$\text{Sample mean: } \bar{x} = (1/N)\,\Sigma_i\, x_i$$

$$\text{Sample standard deviation: } s = \sqrt{[\Sigma_i (x_i - \bar{x})^2 / (N - 1)]}$$

$$\text{Approximate 95\% CI for the mean: } \bar{x} \pm 1.96(s/\sqrt{N})$$

$$\text{Estimated event probability: } P(X \in A) = (1/N)\,\Sigma_i\, 1\{x_i \in A\}$$

2.2 Convergence as a simulation diagnostic

Because simulation output is itself random, a finite run contains Monte Carlo sampling error. The workbook therefore reports running estimates at **increasing sample sizes**. Convergence is interpreted as stabilization of the estimated output as the number of replications increases. It is important to distinguish numerical convergence from model validity: a stable estimate can still be generated by unrealistic assumptions. This distinction follows established simulation-output practice, which treats confidence intervals and replication size as tools for assessing estimation precision rather than proof that the underlying structural model is correct.

3. Investment Withdrawal Strategy Simulation

3.1 Business problem and decision alternatives

The **investment problem** begins with a fund valued at $1,000,000, allocated equally between a stock portfolio and bonds. The stock component has an assumed annual drift of 8% and volatility of 12%, while the bond component earns 5% per year. The investor withdraws $120,000 at the end of each year. The decision is not the annual withdrawal amount, which is fixed, but the sequence in which the two asset pools are depleted.

Under the **parallel withdrawal strategy**, $60,000 is withdrawn from stocks and $60,000 from bonds each year. If one component cannot supply its $60,000 share, that component is emptied and the shortfall is transferred to the other component. Under the **bond-first strategy**, the annual $120,000 withdrawal is taken from bonds while sufficient bond capital remains; when bonds can no longer cover the withdrawal, the shortage is taken from stocks. The outcome of interest is **fund life**: the number of years until the combined balance is depleted.

3.2 Dynamic model

Let S_t and B_t denote the stock and bond balances before the year-t withdrawal. The stock portfolio is modeled using a discrete-time geometric Brownian motion update. If Z_t is a standard Normal shock, the gross stochastic growth factor over one annual interval is $\exp[(\mu - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t}\, Z_t]$. This formulation makes stock returns multiplicative and ensures that uncertainty compounds through time. The bond account evolves deterministically at the fixed annual rate after withdrawals.

$$\textit{Stock growth factor} = \exp[(\mu - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t}\, Z_t], \quad Z_t \sim N(0,1)$$

For the parallel strategy, shortage variables measure whether either asset class is unable to provide its scheduled $60,000 withdrawal. The shortage is transferred to the other asset class. For the bond-first strategy, the bond account absorbs the $120,000 annual withdrawal until its balance is insufficient; the remaining amount is then deducted from stocks. In both cases, an alive indicator equals one while the combined fund remains positive. Summing this indicator across the simulated path yields the stopping time.

3.3 Simulation design and implementation

Parameter	Meaning	Workbook value
B_0	Initial bond fund	$500,000
S_0	Initial stock fund	$500,000
μ	Annual stock drift	8%
σ	Annual stock volatility	12%

r	Annual bond return	5%
Δt	Time step	1 year
Maximum horizon	Maximum simulated path length	50 years
N	Monte Carlo replications per strategy	2,000

Each strategy is simulated over a maximum 50-year path. The workbook generates a stochastic stock return each year, applies the appropriate withdrawal rule, updates stock and bond balances, evaluates whether the fund remains alive, and records the stopping time. The process is then replicated 2,000 times for each strategy. This produces two empirical distributions of fund life that can be compared on expected duration, dispersion, confidence intervals, and distribution shape.

3.4 Results

Metric	Parallel withdrawal	Bond-first withdrawal
Mean fund life	11.9445 years	12.5695 years
Standard deviation	1.5635	2.3004
95% CI for mean	[11.8760, 12.0130]	[12.4687, 12.6703]
Minimum observed	9 years	8 years
Maximum observed	33 years	30 years
Difference in means	—	+0.6250 years

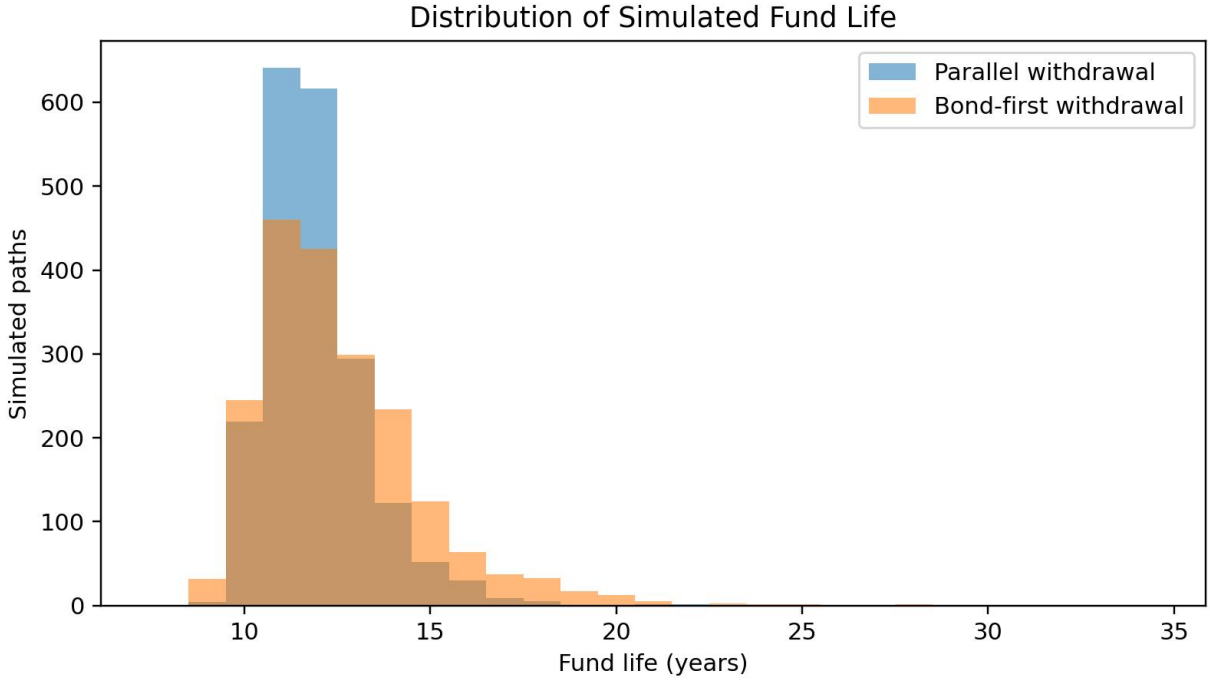


Figure 1. Empirical distribution of fund life under the two withdrawal strategies. Source: project workbook simulation output.

The **bond-first strategy** increases estimated mean fund life from 11.9445 to **12.5695 years**, an increase of 0.6250 years, or approximately 7.5 months. The 95% confidence interval for the parallel-strategy mean is [11.8760, 12.0130], while the corresponding interval for bond-first is [12.4687, 12.6703]. Within this simulation design, the separation between these intervals supports the interpretation that the observed mean difference is not merely a small fluctuation in the reported Monte Carlo sample.

The distributional comparison is more nuanced than the mean alone. The bond-first strategy has a **larger standard deviation** (2.3004 versus 1.5635), and its observed minimum is 8 years compared with 9 years for the parallel strategy. At the same time, the parallel strategy records the larger sample maximum (33 versus 30 years). **Thus, the bond-first policy shifts the center of the distribution to the right but does not dominate the parallel policy in every simulated path**. The policy comparison is therefore best understood as a distributional trade-off rather than a deterministic guarantee.

3.5 Convergence and simulation precision

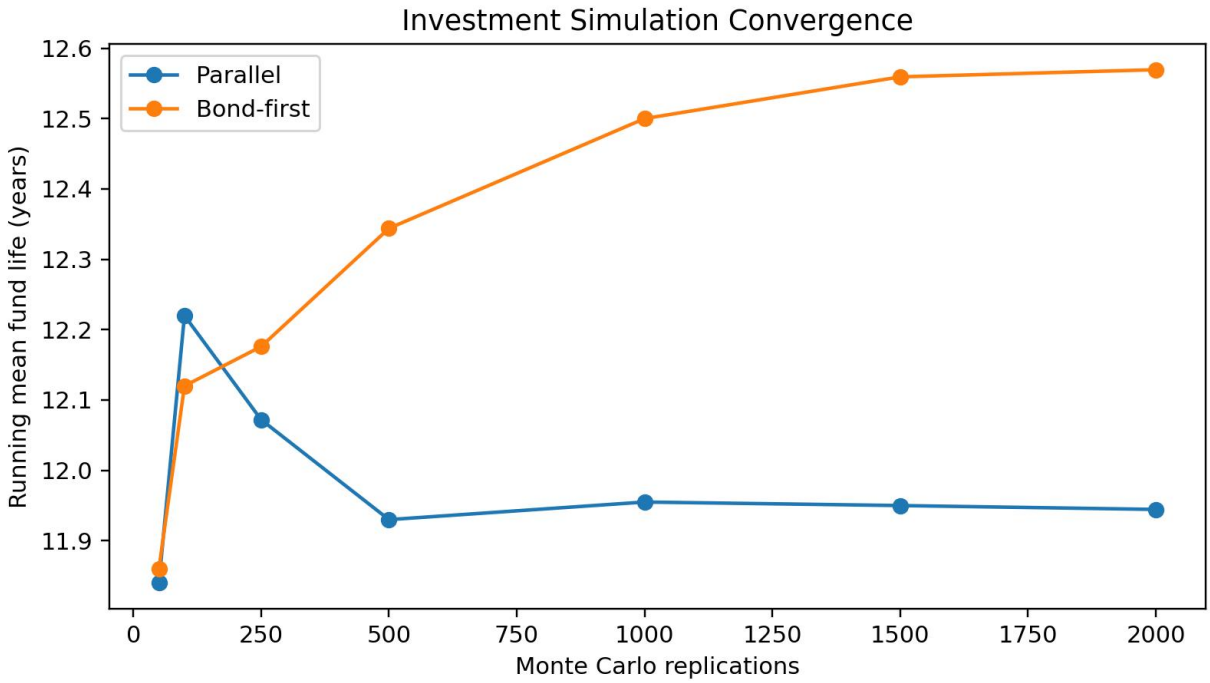


Figure 2. Running mean fund life as the number of Monte Carlo replications increases. Source: project workbook convergence analysis.

The convergence analysis is particularly informative because it shows why a policy conclusion should not be drawn from a small simulation sample. At 50 replications, the running means are 11.84 years for parallel withdrawal and 11.86 years for bond-first, a difference of only 0.02 years. At 100 replications, the ordering temporarily reverses: 12.22 years for parallel versus 12.12 years for bond-first. As the simulation grows, the bond-first advantage becomes increasingly stable: the difference is 0.414 years at 500 replications, 0.545 years at 1,000, 0.6093 years at 1,500, and 0.625 years at 2,000.

This pattern illustrates Monte Carlo sampling error directly. Small samples can generate noisy policy comparisons, including a temporary reversal of the final ordering. By 2,000 paths, the running means have stabilized sufficiently for the workbook's final comparison to be interpreted with substantially greater confidence.

3.6 Managerial interpretation

The economic mechanism behind the bond-first result is the sequencing of exposure to the higher-growth but stochastic asset. Withdrawing bonds first leaves a larger share of stock capital invested for longer, allowing more time for the assumed 8% stock drift to compound. The parallel strategy continuously removes capital

from both asset pools, reducing stock exposure earlier in the fund's life. Under the workbook's assumptions, preserving stock exposure raises expected fund longevity.

However, the larger standard deviation under bond-first also matters. A decision maker concerned only with expected longevity may prefer the higher simulated mean, while a decision maker sensitive to dispersion or early-depletion outcomes should examine the entire distribution. The simulation therefore reframes the managerial question from 'Which rule lasts longer in a typical path?' to 'How does each withdrawal policy reshape the distribution of longevity under return uncertainty?'

4. Insurance Loss-Ratio Risk Simulation

4.1 Business problem

The insurance model evaluates **annual performance** for a sales agent whose compensation depends on the **loss ratio**. Annual performance is uncertain because the number of policies sold varies, policy coverage differs across customers, only some customers purchase excess liability, claims occur probabilistically, and claim severities are random. The key output is the **annual loss ratio R**, defined as total losses divided by total premiums. An annual ratio above one is treated as an adverse performance event because losses exceed premiums.

$$R = TL / TP, \text{ where } TL = \text{total annual loss and } TP = \text{total annual premium}$$

4.2 Stochastic inputs and random sample size

Component	Workbook assumption	Role in simulation
Annual policy sales S	Normal with mean 100 and SD 10	Random annual portfolio size
Basic liability X	Uniform [50, 200]	Heterogeneous policy size
Excess coverage	Purchased with probability 0.40; adds 0.2X	Coverage-choice uncertainty
Claim occurrence	Bernoulli with probability p_c	Claim-frequency uncertainty
Claim severity	Exponential with mean r_c × premium	Loss-severity uncertainty
Simulation horizon	500 years	Monte Carlo output sample

Random annual sales create a variable sample-size problem. The spreadsheet handles this by creating a fixed capacity of 200 potential policies and using an indicator, pick = 1(ID ≤ S), to select the policies that are active in a given simulated year. This design keeps the worksheet dimensions fixed while allowing the realized number of policies to vary from year to year. Real premium and real loss are obtained by multiplying potential-policy values by the selection indicator.

The scenario design separates claim frequency from severity. Four (r_c, p_c) combinations are specified: (5, 0.16), (10, 0.08), (20, 0.04), and (80, 0.01). In each case r_c × p_c = 0.80. Holding this product constant provides a controlled way to study why similar expected loss intensity can coexist with very different variability and tail behavior. The final saved 500-year output analyzed in the workbook corresponds to the active model output shown in the insurance calculation sheet and analysis sheet.

4.3 Simulation procedure

For each simulated year, the model first generates annual sales. It then generates up to 200 potential basic liabilities, excess-coverage choices, premiums, claim indicators, and exponential claim severities. The pick indicator retains only the first S policies, producing realized annual premium and loss vectors. These are

aggregated to total premium and total loss, from which the annual loss ratio is calculated. The process is repeated for 500 years, creating an empirical distribution of R and an indicator for whether R exceeds one.

4.4 Results

Metric	Saved simulation result
Mean loss ratio	0.836729
Standard deviation	0.890498
Minimum observed R	0.000656
Maximum observed R	7.391006
P(R > 1)	29.4%
Standard error of mean	0.039824

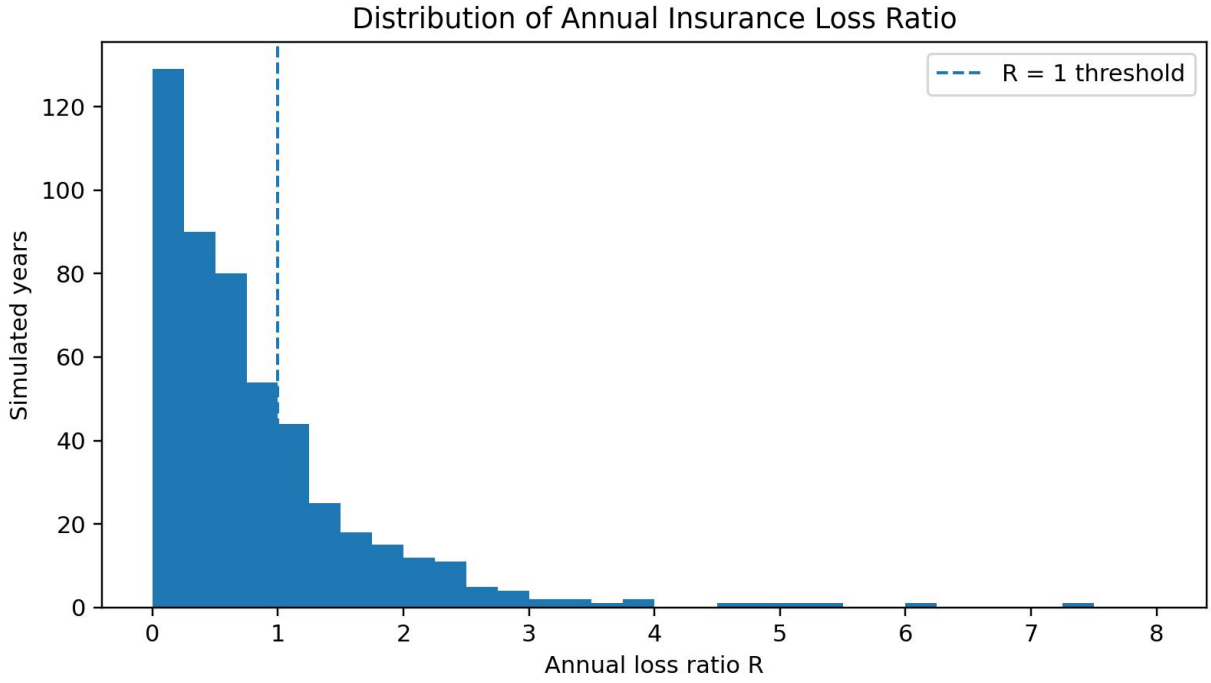


Figure 3. Empirical distribution of annual insurance loss ratios across 500 simulated years. The dashed line marks R = 1. Source: project workbook simulation output.

The 500-year simulation produces a mean annual loss ratio of **0.8367**. The standard deviation, 0.8905, is slightly larger than the mean, indicating substantial year-to-year dispersion. The observed ratios range from 0.0007 to 7.3910, and **29.4%** of simulated years have R > 1. This tail probability is directly decision-relevant: **in nearly three out of ten simulated years under the saved model output, aggregate losses exceed aggregate premiums.**

The maximum observed loss ratio is far above both the mean and the threshold of one, revealing a pronounced **right tail**. This shape is consistent with the model's claim-severity mechanism: exponential losses permit occasional large realizations, and aggregate annual performance can be dominated by a small number of severe claims. Consequently, the mean loss ratio alone is not a sufficient description of performance risk.

4.5 Convergence of insurance estimates

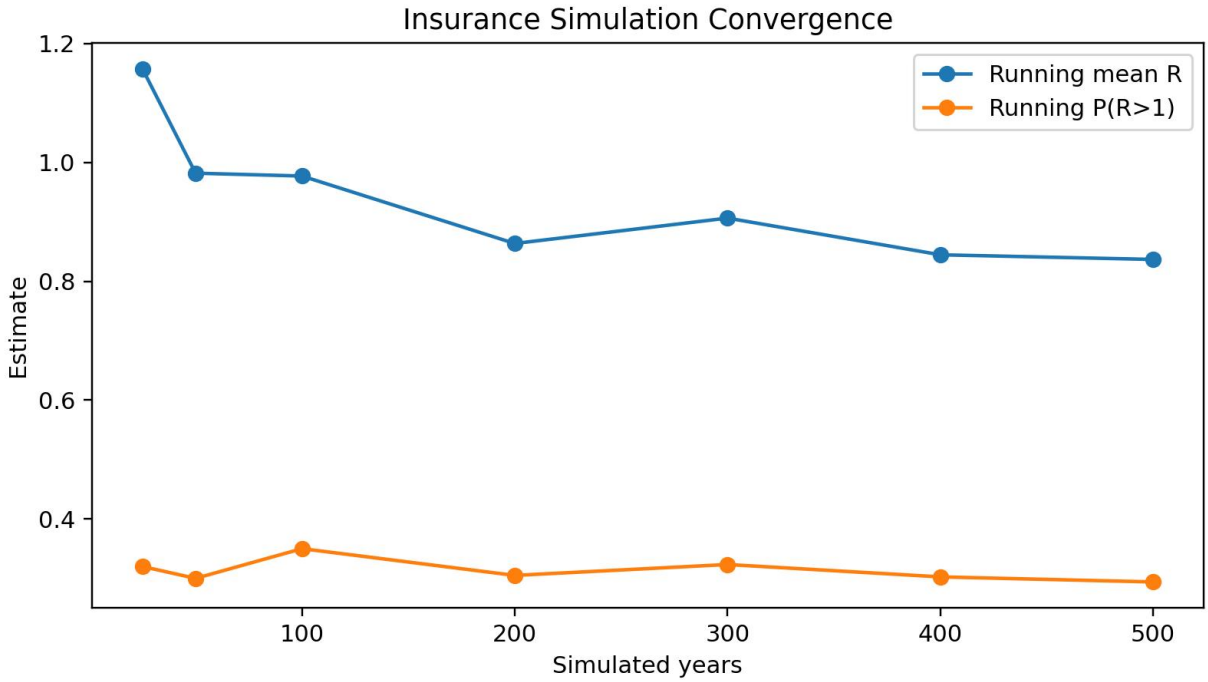


Figure 4. Running estimates of the mean loss ratio and P(R > 1) as simulated years accumulate. Source: project workbook convergence analysis.

The insurance convergence path shows substantial small-sample variation. After 25 simulated years, the running mean loss ratio is 1.1577 and P(R > 1) is 32.0%. At 50 years, the mean falls to 0.9819; at 100 years it is 0.9771, while the estimated exceedance probability rises to 35.0%. With additional replications, the estimates move toward the final 500-year values: the mean is 0.8636 at 200 years, 0.9063 at 300, 0.8446 at 400, and 0.8367 at 500. The corresponding exceedance probability ends at 29.4%.

The non-monotonic convergence path is substantively meaningful. Rare high-severity claims can materially shift a running mean when the sample is small, so additional replications are especially important for tail-sensitive insurance models. The final standard error of the mean, approximately 0.0398, quantifies the remaining Monte Carlo uncertainty around the reported average.

4.6 Business implications

The insurance simulation demonstrates why performance management should separate expected outcome from risk dispersion. An average loss ratio below one does not imply that adverse years are rare: the saved simulation simultaneously reports a mean below one and a 29.4% probability of exceeding one. For compensation design, underwriting oversight, or risk budgeting, the exceedance probability and upper tail may therefore be as important as the mean.

The frequency-severity scenario structure provides a second managerial lesson. When expected loss intensity is held approximately constant through r_c × p_c, shifting risk from frequent/moderate claims toward rare/severe claims can leave the expected value broadly similar while increasing volatility and right-tail exposure. This distinction is important whenever a business metric is evaluated against a threshold: two environments with similar averages can produce very different probabilities of unacceptable outcomes.

5. Integrated Discussion

5.1 Two different forms of uncertainty

The two models illustrate complementary simulation architectures. The investment model is **path-dependent**: today's withdrawal changes tomorrow's state, and random returns compound over time. The insurance model is **cross-sectional** within each year and then replicated across years: many policy-level random variables aggregate into an annual performance measure. These differences explain why the most useful outputs also differ. Stopping-time distributions and policy comparisons are central to the investment problem; loss-ratio dispersion and threshold probabilities are central to the insurance problem.

5.2 Why distributional analysis changes the decision

Both studies show the limitations of mean-only reasoning. In the investment model, bond-first has the higher expected fund life, but also higher dispersion and a lower observed minimum. In the insurance model, the mean loss ratio is below one, yet the probability of exceeding one remains economically material and the distribution contains a long upper tail. Simulation makes these distinctions visible because it preserves the distribution of outcomes rather than collapsing uncertainty into a deterministic forecast.

5.3 Spreadsheet modeling as an analytical tool

The spreadsheet implementation is valuable not merely because it generates random numbers, but because the model is transparent. Inputs, state transitions, random-variable generation, aggregation, output statistics, distributions, and convergence diagnostics can be traced directly from cells and formulas. This transparency supports model auditing and managerial communication, while also demonstrating how statistical reasoning and business logic can be integrated in a reproducible analytical workflow.

6. Limitations

The results are conditional on the model assumptions and should not be interpreted as forecasts of an actual investment fund or insurance portfolio. The investment model assumes constant stock drift and volatility, normally distributed annual shocks within a geometric Brownian motion framework, a fixed 5% bond return, annual time steps, fixed nominal withdrawals, and no transaction costs, taxes, fees, inflation, or regime changes. Alternative return dynamics or adaptive withdrawal rules could materially alter the distribution of fund life.

The insurance model assumes independent policy outcomes, Normal annual sales, Uniform basic liability, a fixed excess-coverage probability, Bernoulli claim occurrence, and exponential claim severity. It does not model dependence among claims, heterogeneous underwriting classes, deductibles, reinsurance, inflation, pricing adaptation, or multi-year learning. The 200-policy capacity is an implementation device rather than an estimated operational constraint.

Finally, both studies contain finite-simulation error. The convergence analysis and confidence intervals address numerical precision, but they do not validate the structural assumptions. In simulation research, model validation and output precision are distinct concerns: increasing the number of replications can reduce Monte Carlo error without correcting a misspecified model.

7. Conclusion

This portfolio project uses Monte Carlo simulation to evaluate two **financial decisions** whose consequences cannot be understood adequately from deterministic averages alone. In the **investment model**, preserving stock exposure through a **bond-first withdrawal policy** raises simulated mean fund life by approximately **0.625 years** relative to parallel withdrawals under the specified assumptions, although the bond-first distribution is also more **dispersed**. In the **insurance model**, the saved 500-year simulation produces a **mean loss ratio** of approximately **0.837** but an estimated **29.4%** probability that annual losses exceed premiums, highlighting the importance of tail risk.

Methodologically, the project demonstrates a **complete simulation workflow**: translating business problems into stochastic models, generating random variables, constructing dynamic or aggregate outcomes, replicating the model, summarizing output distributions, checking convergence, and translating statistical results into managerial implications. The central analytical lesson is that **uncertainty should be evaluated through distributions, not point estimates alone**. Expected value, variability, tail probability, and simulation precision jointly determine the quality of a decision under uncertainty.

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